

- 5 (a) ohm = volt / ampere B1 [1]
- (b) $\rho = RA / l$ or unit is Ωm C1
 units: $\text{VA}^{-1}\text{m}^2\text{m}^{-1} = \text{NmC}^{-1}\text{A}^{-1}\text{m}^2\text{m}^{-1}$ C1
 $= \text{kgm}^2\text{s}^{-2}\text{A}^{-1}\text{s}^{-1}\text{A}^{-1}\text{m}^2\text{m}^{-1}$
 $= \text{kgm}^3\text{s}^{-3}\text{A}^{-2}$ A1 [3]
- (c) (i) $\rho = [3.4 \times 1.3 \times 10^{-7}] / 0.9$ C1
 $= 4.9 \times 10^{-7} (\Omega\text{m})$ A1 [2]
- (ii) max = 2.(0) V A1
 min = $2 \times (3.4 / 1503.4) = 4.5 \times 10^{-3} \text{ V}$ A1 [2]
- (iii) $P = V^2 / R$ or $P = VI$ and $V = IR$ C1
 $= (2)^2 / 3.4$
 $= 1.18$ (allow 1.2) W A1 [2]
- (d) (i) power in Q is zero when $R = 0$ B1 [1]
- (ii) power in Q = 0 / tends to zero as $R = \text{infinity}$ B1 [1]

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6 (a) extension is proportional to force (for small extensions)

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